

GC University Lahore

BS Computer Science

Session 2021-25

Examination: Final 2024

Course Code: CS-4217

Time allowed: 155 minutes

Semester: 6th / 8th

Title: Mobile Application

Marks: 40

Subjective

Attempt any Four Questions

Question No. 2: [Marks 5+5]

a. Sketch a diagram illustrating the lifecycle of a Stateful Widget in Flutter, focusing on the execution flow of `initState`, `build`, `setState`, and `dispose` methods.

Alternatively, create a diagram representing the Android activity lifecycle.

b. What will be the output of the Dart code given below?

```
class Algebra {
  void output() {
    print("Hello Algebra");
  }
}

class LinearAlgebra extends Algebra {
  @override
  void output() {
    print("Hello Linear Algebra");
  }
}

void main() {
```

```
late Algebra obj;  
Algebra obj1 = Algebra();  
obj1.output();  
LinearAlgebra obj2 = LinearAlgebra();  
obj2.output(); obj1 = obj2;  
obj = obj1; obj.output();  
obj2.output();  
}
```

Question No. 3: [Marks 10]

As a developer transitioning to Kotlin for mobile app development, how do Kotlin's features like concise syntax, extension functions, coroutines, and interoperability with Java improve the development process, code safety, and runtime error prevention?

Question No. 4: [Marks 10]

What are some advantages and disadvantages of Kotlin and Flutter?

Question No. 5: [Marks 10]

Write a program to calculate and display the distance between two cities in meters, feet, and inches, with the initial distance given in kilometers. Implement error handling for invalid inputs using appropriate methods. The results should be displayed in the interface. You may use Kotlin or Flutter; the syntax does not need to be perfect but should provide the basic structure.

Question No. 6: [Marks 10]

You're building a note-taking app and plan to use SQL or Firebase for data management. Outline the process for setting up a database, including creating a table for notes, and describe how to perform CRUD operations with SQL/Firebase statements.

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